

Discovering Deep-Tier Suppliers: LLM-Driven Manufacturing Site Inference for Enhanced Visibility

Keita Mizushina

Hitachi, Ltd.

Research & Development Group

Kokubunji, Tokyo, Japan

keita.mizushina.ez@hitachi.com

Mika Takata

Hitachi, Ltd.

Research & Development Group

Kokubunji, Tokyo, Japan

mika.takata.nc@hitachi.com

Tsunehiko Baba

Hitachi, Ltd.

Research & Development Group

Kokubunji, Tokyo, Japan

tsunehiko.baba.gc@hitachi.com

Ken Naono

Hitachi, Ltd.

Research & Development Group

Kokubunji, Tokyo, Japan

ken.naono.aw@hitachi.com

Abstract

The critical challenge of identifying deep-tier manufacturing sites in complex global supply chains hinders effective risk management. This paper proposes an LLM-driven Manufacturing Site Inference Technology to improve this visibility. Our novel approach employs a fine-tuned Large Language model to process diverse, unstructured, and sometimes contradictory data sources. It infers factory locations at the address level, which are then converted into latitude and longitude coordinates. The system uniquely leverages International Organization for Standardization (ISO) 9001 certifications, extracted via specialized prompt engineering, and is fine-tuned with domain-specific data including Harmonized System (HS) codes to enhance its accuracy. Validated against real-world procurement data, the technology achieves accuracy up to 86.7% in identifying manufacturing sites, facilitating a human-in-the-loop approach for verification. This provides tangible, site-specific visibility, enabling more effective risk management and enhancing supply chain resilience.

CCS Concepts

• Applied computing → Supply chain management.

Keywords

Supply chain, Large Language Models, Manufacturing site, Risk management, Geopolitical risk

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1 Introduction

Global supply chains are vulnerable to geopolitical shifts and trade tensions [13], making the accurate identification of all parts' manufacturing locations a critical capability for modern enterprises. The specific manufacturing sites of components are essential not only for operational efficiency but also for mitigating significant risks that can arise from geopolitical events [10]. An over-reliance on specific regions can expose companies to procurement disruptions due to political instability or trade restrictions, with recent analyses highlighting challenges in tracing conflict minerals [11] and managing risks in multi-tier supply chains [12]. Furthermore, an inability to trace the origins of materials can lead to ethical and compliance issues, particularly concerning Environmental, Social, and Governance (ESG) standards, which have become increasingly important in corporate governance and consumer expectations [9, 16]. The lack of visibility into deep-tier supplier (Tier 2 supplier or beyond), often where critical components originate, can result in unexpected shortages and create severe vulnerabilities, as evidenced by recent challenges in the semiconductor industry [6, 21]. Thus, the need for improved resilience and adaptability in supply chains is paramount, underscoring the necessity for novel solutions that enhance visibility across all tiers [15, 19].

However, achieving this comprehensive visibility remains a significant challenge. Traditional methods for mapping supply chains, such as manual surveys and direct inquiries with suppliers, have proven increasingly inadequate in today's complex environments [23, 24]. These approaches are not only time-consuming and resource-intensive but are also complicated by the reluctance of suppliers to disclose manufacturing locations due to competitive pressures [5]. While recent LLM-driven approaches have shown promise in extracting buyer-supplier relationships [3] or predicting hidden network links [24], they often stop short of identifying the specific, physical manufacturing sites that are crucial for granular, geospatial risk assessment.

To address this critical gap, this paper introduces a novel LLM-driven technology for automatically inferring manufacturing locations by leveraging the advanced capabilities of Large Language Models. Our approach processes diverse, unstructured, and often contradictory data sources—including parts supplier lists, public



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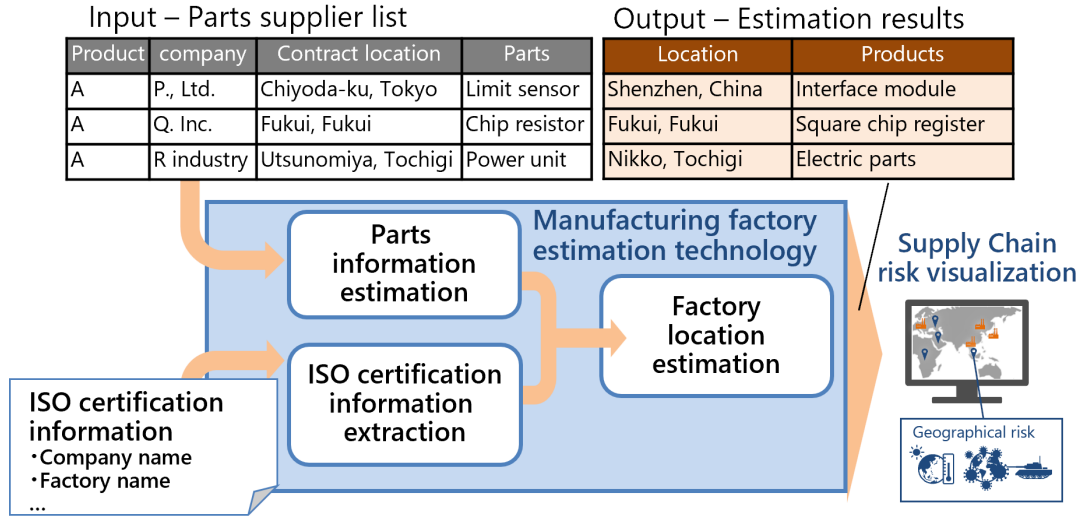


Figure 1: Overview of Proposed method. Combine multiple data sources to estimate factory locations and compare them with geopolitical risks.

web information, and ISO 9001 certifications—to infer precise factory locations at the address level, which are then converted to latitude/longitude coordinates. The primary contributions of this study are threefold: (i) This study develops a novel LLM-driven inference pipeline, utilizing a fine-tuned LLM, to discover deep-tier manufacturing sites, thereby enabling the visualization of previously hidden supply chain risks. (ii) This study also establishes a robust methodology that is resilient to conflicting and incomplete information, ensuring reliable outputs even when dealing with the complexities of real-world data. (iii) This study demonstrates the practical viability of our technology by achieving over 80.1% estimation accuracy on real-world procurement data, validating its potential for significant operational impact.

This paper is structured as follows. Section 2 reviews related work to position our study within the existing literature. Section 3 details the proposed methodology. Section 4 presents the experimental setup and evaluation results. Section 5 provides a multi-faceted discussion of the findings, and Section 6 concludes the study.

2 Related Work

This section surveys the existing technological landscape and academic literature to highlight the persistent challenges in achieving deep-tier visibility that motivate our work.

Achieving supply chain visibility, particularly in the deep tiers (Tier 2 and beyond), is a persistent challenge. Traditional supply chain management systems, such as enterprise resource planning (ERP), are foundational but primarily focus on managing transactional data with direct, Tier 1 suppliers [5]. While these systems provide essential operational support, they lack the capability to uncover the complex, multi-layered networks in deeper tiers [14, 18]. This limitation hampers the ability to assess geographical risk concentrations and develop effective resilience strategies. A survey of Chief Procurement Officers highlighted this gap, revealing that

while 70% felt confident about their Tier 1 visibility, this figure plummeted to just 15% for Tier 2 and beyond [10].

In response to these limitations, emerging technologies such as blockchain and the Internet of Things (IoT) have been proposed to enhance traceability and transparency [25]. Blockchain technology represents a promising approach to enhancing supply chain traceability by providing immutable transaction records [17]. However, its widespread implementation faces challenges, such as varying levels of technological adoption among suppliers and the need for industry-wide standards. Similarly, IoT devices offer significant potential for real-time tracking of goods, contributing to improved transparency in supply chains [1]. Nonetheless, their deployment requires substantial investment and poses challenges in scalability and data security, which may limit feasibility for certain stakeholders.

Recent advancements in Artificial Intelligence (AI), particularly in Natural Language Processing (NLP) and Large Language Models (LLMs), offer new opportunities to address these challenges. AI-driven methods have demonstrated their ability to extract valuable insights from large volumes of unstructured text data available publicly [2, 22, 26, 27]. For example, one study proposed a deep neural network-based methodology to extract buyer-supplier relationships from news articles [24]. While effective in mapping relational networks, this approach does not address the challenge of identifying specific manufacturing locations. Similarly, the Supply Network Link Predictor (SNLP) [4] infers hidden supplier interdependencies using network topology, which addresses incomplete supply chain knowledge but remains focused on relational links rather than physical site identification.

Despite these advancements, a review of the literature reveals an important research gap. While traditional systems provide foundational support for Tier 1 visibility, and newer technologies such as

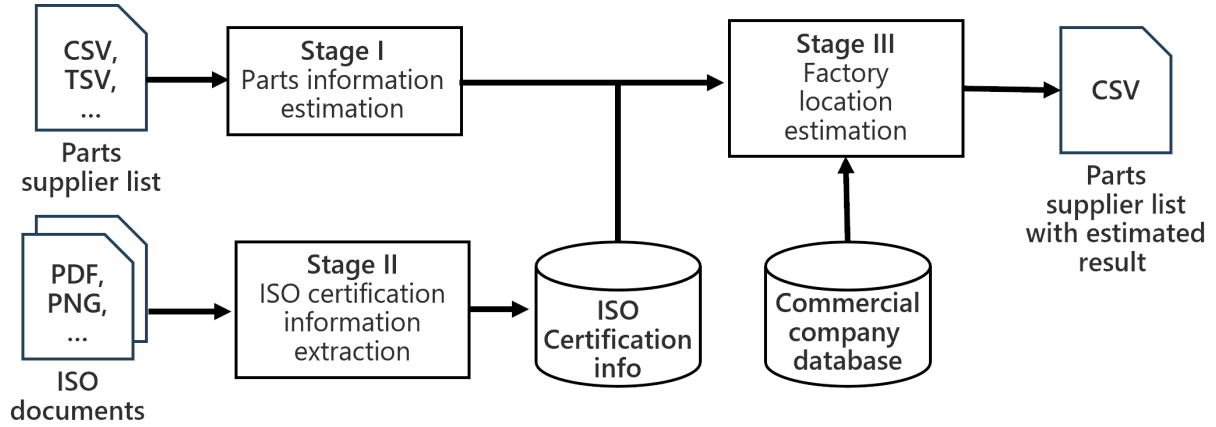


Figure 2: Block diagram of factory location estimation method.

Table 1: Example of parts supplier list.

Product name	Company name	Location	Part name	Model number	Tier
Product A	Company B	Chiyoda-ku, Tokyo	Cable	00055A	1
Product A	Company C	Fukui, Fukui	Steel plate	E-00094	1
Product A	Company D	Utsunomiya, Tochigi	Screw bolt	SC-08056-1	1

[Scope]

1. Manufacturing of products
2. Sales of products
3. Maintenance and inspection of products
4. Head office functions

A office : Nagoya, Aichi, Japan

[Function 4.]

B factory : Nagaoka, Niigata, Japan

[Function 1. and 3.]

...

Figure 3: Example of ISO 9001 certification data. Crucially, this data requires accurate extraction of the correspondence between a facility (e.g., 'B factory') and its designated scope of certification (e.g., functions '1. Manufacturing' and '3. Maintenance')."

blockchain and IoT offer novel solutions for traceability, their adoption challenges limit their applicability across deep-tier networks. Similarly, state-of-the-art AI applications, such as those focusing on relational mapping or network topology, have made significant strides in addressing supply chain complexities. However, the automated identification of specific, physical manufacturing sites remains an unaddressed challenge. This capability is essential for

granular geospatial risk assessments and proactive supply chain management, highlighting the gap our research seeks to bridge.

3 Proposed Method: LLM-Driven Manufacturing Site Inference

3.1 System Overview and Architecture

To address the research gap identified in Section 2, this paper proposes a novel methodology leveraging Large Language Models (LLMs) to automatically infer manufacturing sites from diverse and unstructured data sources. The method is structured as a multi-stage pipeline, consisting of three primary stages: Parts Information Estimation, ISO Certification Information Extraction, and Factory Location Estimation. Figure 1 illustrates the conceptual overview, highlighting the pipeline's overall flow, while Figure 2 provides a detailed view of the system architecture, including data input, processing, and output stages.

The first stage, Parts Information Estimation, takes a parts supplier list as its input. A fine-tuned LLM processes this list to normalize part names and classify their corresponding product sectors. The output is an enriched supplier list containing standardized parts and sector information.

The second stage is ISO Certification Information Extraction. This stage uses ISO 9001 certification documents, often in unstructured PDF or image formats, as input. A specialized LLM extracts key details into a structured format. These details include the company name, factory name, factory address, and the scope of certification.

The final stage is Factory Location Estimation. It integrates the structured outputs from the preceding stages with data from external commercial databases. The core LLM then processes this consolidated data to infer the final manufacturing address for each part. The system also converts the address to latitude and longitude coordinates. The final output enables geographical risk visualization and analysis. Figure 2 shows the architecture of proposed methods.

3.2 Data Acquisition and Pre-processing

The effectiveness of our method hinges on two primary data sources: parts supplier lists and ISO 9001 certifications.

Parts supplier lists are provided by procurement officers and detail which parts are procured from which companies for a specific product. A critical limitation of this data is that the address listed is typically the supplier's head office or a regional sales office, not the actual manufacturing site. Our goal is to identify the factory location, which is essential for risk assessment.

To overcome this limitation, we utilize ISO 9001 certification documents [20]. ISO 9001 is a globally recognized standard for quality management systems and provides certification at the factory level [8]. This data is highly valuable because it explicitly states the name, location, and certified functions (e.g., "manufacturing," "design") of each factory. However, these documents are typically unstructured, requiring a robust information extraction process. A key challenge is correctly associating a factory's designated scope of certification with its textual description, especially when they appear in different sections of a document, as illustrated in Figure 3.

3.3 Core LLM: Fine-Tuning and Prompt Engineering

At the heart of our technology is a Large Language Model customized for supply chain analysis. We selected Meta's Llama 3 based model [7] as the foundational model due to its strong reasoning capabilities and adaptability to domain-specific fine-tuning.

To specialize the model, we performed a rigorous fine-tuning process. We utilized a proprietary dataset of approximately 5,000–10,000 anonymized entries structured in JSON format. Each entry contains domain-specific knowledge, mapping supplier details, part characteristics, and verified manufacturing site information. A key component of this dataset is the integration of Harmonized System (HS) codes. Associating parts with their HS codes helps the model learn the correlation between part characteristics and typical manufacturing regions or required factory capabilities. This domain-specific fine-tuning significantly enhances the LLM's ability to identify subtle patterns indicative of a manufacturing presence within unstructured text.

For the task of extracting information from semi-structured ISO 9001 documents, we employ specific prompt engineering strategies. We primarily leverage few-shot prompting to guide the model's output. This technique involves providing the model with a small number of illustrative examples within the prompt itself. For instance, a prompt might include a text snippet from a sample ISO document and the corresponding desired structured JSON output.

By seeing these examples, the LLM learns the precise task requirements, including how to handle complex layouts and produce consistently formatted data. This approach significantly improves data extraction accuracy from new and varied certification documents.

3.4 Multi-Source Inference, Validation, and Geocoding

The final stage of our methodology synthesizes all collected and extracted information to identify the most probable manufacturing site for each part. This culminating process leverages our fine-tuned Large Language Model to meticulously navigate and interpret the complex, often incomplete or disparate, information characteristic of deep-tier supply chains. The process consists of four key steps: data aggregation and conflict resolution, textual address inference, geocoding, and validation with supplementary data.

First, the LLM aggregates data from all preceding stages and navigates the potentially diverse and sometimes conflicting information. For instance, a supplier may have different addresses listed across a contract document and an ISO certificate. To resolve such discrepancies, the LLM employs its advanced contextual understanding and the pre-assigned reliability of data sources (e.g., ISO 9001 certifications are weighted higher for factory-specific details than general corporate websites). Our fine-tuned LLM serves as the core reasoning engine for resolving data discrepancies. The process is guided by a structured prompt designed for evidence-based inference. When conflicting information (e.g., the headquarters address listed in the contract and the factory address listed in the ISO certificate) is presented, the LLM is instructed to autonomously retrieve information from the web and prioritize sources that are likely to indicate the physical manufacturing facility.

An example of a simplified version of the conflict resolution prompt is shown below: "Infer the manufacturing address for [Part X] from [Supplier Y]. Evidence: A) Contract Address: [Address A], B) ISO 9001 Certified Factory: [Address B, Scope: Manufacturing]. C) WEB search results: Address listed on the official website: [Address B]. Justify your choice by prioritizing the evidence directly related to manufacturing activities." However, it has been experimentally revealed that prompt engineering strategies are insufficient. This is because prompt engineering alone cannot address situations where manufacturers do not disclose where or what they manufacture.

Our investigation revealed that POs possess implicit knowledge—such as the general region where a plant is located—that can help identify specific factories. A key conceptual goal of our methodology is to eventually capture this knowledge through a human-in-the-loop feedback system. The core concept envisions an iterative process where the system's estimations are reviewed by a PO. This feedback would then be collected and used to refine future inferences, allowing the model to learn user-specific supplier selection trends over time. Such a system would leverage the LLM's contextual understanding to efficiently gather and integrate expert knowledge, progressively improving its accuracy. While the full implementation and evaluation of this continuous learning loop are designated as future work, the system presented in this paper serves as the foundational first step toward this vision. The current model's primary inference is based on the data aggregation and conflict resolution mechanisms described earlier.

Table 2: Experimental environment.

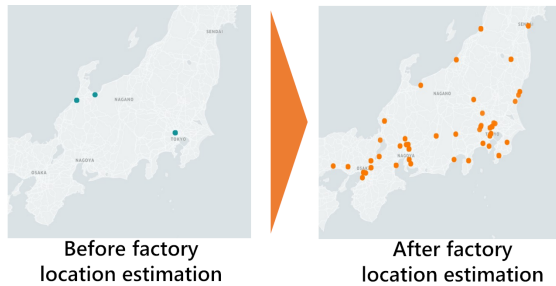
Component	Specification
OS	Ubuntu 22.04
CPU	Intel Xeon W7-3445
GPU	2x NVIDIA RTX A4500

Table 3: Dataset specifications.

Attribute	Description
Data Source	Procurement Records (BOMs)
Data Format	Tabular Data (CSV-based)
Total Parts	3,706

Table 4: Quantitative results of parts manufacturing factory estimation.

Product	Category	Total Parts	Estimated Parts	Correctly Estimated
Product A	Inspection Machine	1374	1066 (77.6%)	924 (86.7%)
Product B	Manufacturing Machine	2034	2006 (98.6%)	1629 (80.1%)
Product C	Surface processing machine	298	263 (88.3%)	248 (83.1%)

**Figure 4: Manufacturing site visibility: 'Before factory location estimation' (left) shows sparse initial data, while 'After' (right) reveals significantly more sites identified by our LLM system.**

Second, based on this conflict resolution, the system deduces the most probable textual address of the manufacturing facility. The primary output of this step is a reasoned textual address, such as a plausible street address or a combination of a well-recognized factory name and its locality, not yet geospatial coordinates.

Third, the inferred textual address is systematically converted into precise latitude and longitude coordinates. This is achieved through the robust application of an external geocoding API, ensuring a standardized and geographically accurate representation of the inferred site.

Finally, to enhance reliability, each inferred location is cross-referenced with a variety of supplementary contextual data sources. These diverse sources include official corporate websites detailing factory locations, relevant news articles or industry reports, and data from online map services and business directories. In cases where primary sources are insufficient or conflicting even after the LLM's initial inference, our process incorporates data from established commercial company databases to supplement or verify location details. Any inferences with unresolved discrepancies after this validation process are flagged for a final human-in-the-loop review. This systematic procedure ensures that the final output—a list of manufacturing sites with their specific geospatial coordinates—is as accurate and comprehensive as possible, forming a robust basis for subsequent analysis and risk assessment.

4 Experiment and Result

4.1 Experimental Setup and Dataset

This study evaluated the proposed system using a real-world dataset comprising procurement records for three distinct manufacturing products, covering over 1000 parts. Ground truth data was established through meticulous manual verification by experienced procurement personnel. The data set comprised actual Bills of Materials (BOMs) from our internal manufacturing operations, reflecting a mix of global and Japan-centric suppliers. To create a reliable ground truth for evaluation, our procurement specialists meticulously contacted suppliers to verify the actual manufacturing locations for each part. The computational environment in which these experiments were conducted is detailed in Table 2. Furthermore, key characteristics of the input dataset, including file format and dimensions, are summarized in Table 3.

While the dataset predominantly covers mechanical and semiconductor components, it provides a realistic representation of typical manufacturing supply chains. However, it is important to acknowledge that because the dataset used in this study is primarily composed of mechanical and semiconductor components, the direct applicability of our current model and findings to other industrial sectors, such as those involving chemical products, pharmaceuticals, or textiles, which may possess distinct part characteristics, data availability nuances, and unique supply chain structures, warrants further investigation and stands as a subject for future validation research. An overview of the estimation results achieved by our system for the evaluated product categories is presented in Table 4, providing a preliminary indication of its performance. A comprehensive analysis and discussion of these experimental outcomes will follow in Section 4.2.

To assess the proposed method's performance, it is crucial to understand the inherent challenges within our real-world dataset. Analysis of the ground truth revealed that for over 78% of the parts, the contract address was a non-manufacturing site such as a head office or sales branch or blank. Furthermore, approximately 35% of the suppliers were Small-to-Medium Enterprises (SMEs) with a limited digital footprint, lacking detailed websites or public ISO certifications, making inference particularly challenging. These characteristics underscore that the task extends far beyond simple name matching and requires a robust inference capability to achieve high accuracy.

Table 5: Qualitative examples of inference failure scenarios.

Input Part	Input Supplier (Contract)	System Inference (Output)	Ground Truth	Failure Type / Reason
Motor	F Inc. (undisclosed)	F Inc. Tochigi Plant	F Inc. Kagoshima Plant	Incorrect Plant Selection (Similar capabilities across plants)
Bracket	G Metals (undisclosed)	G Metals Osaka Factory	G Metals (small private factory), Ota-ku, JP	Same-Name Entity Confusion (Lack of public data for the true supplier)
Sensor	H corp. (Nagano, JP)	H corp. Nagano Plant	H corp. Nagano Plant AND H corp. China Plant (BCP Site)	Incomplete Inference (Overseas site was not discovered)

Table 6: Results of investigation into factors contributing to estimated results

Experimental conditions	The percentage of correctly estimated parts (Product A)	The percentage of correctly estimated parts (Product B)	The percentage of correctly estimated parts (Product C)
(1) Generic LLM	53.3%	50.1%	47.5%
(2) Generic LLM with ISO data	64.6%	61.6%	70.4%
(3) Fine-tuned LLM (FT-LLM)	77.5%	72.3%	75.1%
(4) Proposed methods (FT-LLM and ISO data)	86.7%	80.1%	83.1%

4.2 Analysis of Estimated Results and Failure Patterns

Table 4 confirms that our LLM-driven pipeline identifies manufacturing sites with an accuracy of 86.7 % for Product A, 80.1 % for Product B, and 83.1 % for Product C. To better understand the errors, we examined 150 mispredictions and grouped them into three recurrent patterns. Representative examples are provided in Table 5.

Incorrect Plant Selection (42 % of all errors): A supplier operates multiple factories with overlapping HS-code portfolios, and the model selects the wrong one. *Example* (Row 1 in Table 5): the *Tochigi* plant was returned instead of the correct *Kagoshima* plant for a machined shaft in Product A.

Same-Name Entity Confusion (31 %): Distinct companies share identical or nearly identical names; the LLM tends to select the firm with the richer digital footprint (e.g., website, ISO record). This issue is particularly frequent in Product C, where 64 % of such confusions involve SMEs that lack any web presence (Row 2).

Generic-Part Opacity (27 %): Commodity fasteners and electronic passives are often supplied by trading firms that do not manufacture them. Because the trading firm’s address usually refers to a sales office or warehouse, the model must infer an unseen contract manufacturer—a task it rarely resolves correctly. More than 70 % of Product B errors belong to this category; a typical outcome is an *Incomplete Inference* where a secondary factory (e.g., a BCP site) is missed (Row 3).

4.3 Comparison with Baseline Methods

To contextualize the performance of our system, we compared it against baseline methods. The primary baseline involved procurement specialists manually contacting suppliers to verify manufacturing locations, a process that took over three weeks for each of

Product A and Product B and required substantial expertise and effort. While this method achieves high accuracy for Tier 1 suppliers, its applicability to Tier2 and beyond is severely limited due to the lack of direct access to deeper-tier supplier information. Additionally, its reliance on manual effort renders it impractical for large-scale supply chain mapping tasks.

In contrast, our proposed method achieves comparable accuracy (86.7% for Product A, 80.1% for Product B) in under two hours, showcasing a transformative improvement in time efficiency. Beyond Tier 1, our system demonstrates its ability to infer manufacturing sites by utilizing publicly available data and ISO certifications, enabling applicability to multi-tier supply chains.

To further validate the technical efficacy of our approach, we introduced a second, technical baseline: a generic, non-fine-tuned Large Language Model. This baseline utilized the same foundational model as our system but performed the manufacturing site inference task without the domain-specific fine-tuning.

The results showed that the accuracy of the non-fine-tuned LLM fell to between 47.5% to 53.3%, significantly underperforming our fine-tuned model.

This performance gap can be primarily attributed to a lack of domain-specific knowledge. A generic LLM cannot fully comprehend the specific functions or industrial classifications from part names alone. The Harmonized System (HS) codes, which our model learns during fine-tuning, provide structured, hierarchical knowledge about part characteristics. Only by learning this domain-specific data can the model make advanced inferences, such as correlating a certain type of part with a factory likely to possess the requisite manufacturing capabilities.

Therefore, this comparison clearly demonstrates that domain-specific fine-tuning is indispensable for achieving high-accuracy inference in the specialized field of supply chain analysis.

4.4 Ablation Study of the Proposed Method

To quantify the contribution of each component in our pipeline, we conducted an ablation study on the *Products* corpus. Four model variants were compared:

- (1) Generic LLM: The original Llama-3 model without task-specific adaptation.
- (2) Generic LLM with ISO data: The original model augmented at inference time with structured information extracted from ISO data.
- (3) Fine-tuned LLM (FT-LLM): The model fine-tuned on our proprietary dataset containing HS-code annotations, but without access to ISO 9001 data.
- (4) Proposed method (FT-LLM and ISO data): The full pipeline, i.e., the fine-tuned model combined with ISO 9001 information and the associated reasoning logic.

The results (Table 6) reveal a clear performance hierarchy. For Product A, the Generic LLM achieved only 53.3% accuracy, underscoring the need for domain knowledge. Injecting ISO data into the generic model raised accuracy only marginally, suggesting that certificate-level information alone lacks the granularity required for this task. Task-specific fine-tuning alone boosted accuracy to 77.5%, indicating that the model learns to associate part descriptors with manufacturing capabilities from HS-code supervision. Finally, combining fine-tuning with ISO data further increased accuracy to 86.7%, demonstrating the complementary value of authoritative, structured metadata.

The same trend holds for Products B and C: most of the improvement stems from fine-tuning, yet ISO data remain essential for attaining high accuracy.

In summary, the ablation confirms that (i) task-specific fine-tuning is the primary driver of performance, and (ii) high-quality external metadata are still necessary to reach state-of-the-art accuracy.

A finer-grained view emerges when the ablation results are cross-referenced with the failure patterns described in Section 4.2.

Incorrect Plant Selection is substantially mitigated by ISO data. Even a generic, non-tuned LLM benefits from plant-level cues contained in the certificates, as seen from the gap between the first and second rows of Table 6.

Same-Name Entity Confusion is likewise reduced by ISO augmentation, because the certificates explicitly link company names, addresses, and product categories. Fine-tuning alone contributes little here; HS-code supervision improves part semantics but cannot disambiguate homonymous firms.

Incomplete Inference also improves when ISO data are present, since overseas or subcontracted plants invisible with LLM internal knowledge.

These observations strongly support the conclusion that ISO-derived metadata are indispensable for closing the remaining performance gap.

The accuracies reported in Table 6 are preliminary. As discussed in Section 3.4, our system continuously incorporates users' implicit feedback. Each interaction enriches the knowledge base, allowing the LLM to refine its estimates over time; we therefore anticipate further gains after deployment.

The ablation findings, together with the error analysis, provide the foundation for the following discussion section, where we position our approach relative to prior work and outline avenues for future enhancement.

5 Discussion

5.1 Interpretation of Result

The quantitative results demonstrate that our system can identify manufacturing sites with high accuracy (over 80.1%). The variance in performance between Product A (86.7%), Product B (80.1%) and Product C (83.1%) does not necessarily indicate a model deficiency, but rather reflects the different underlying structures of their respective supply chains. Product B contained a higher proportion of generic parts, such as bolts and screws, often supplied by distributors who market them under a private brand without operating their own manufacturing facilities. In these cases, the supplier's address does not correspond to a factory, making inference inherently challenging. In contrast, Product A's components were more specialized, typically sourced from suppliers who are also the manufacturers, leading to higher inference accuracy.

The qualitative analysis of failure cases in Table 5 further illuminates the system's operational characteristics. The incorrect plant selection and same-name entity confusion cases underscore the difficulty of disambiguation when public data is sparse or ambiguous. This is particularly true for small-to-medium-sized enterprises (SMEs). These companies often lack a significant digital presence (e.g., a corporate website or ISO certification) not only due to the costs associated with maintaining them, but also because their business models, which may rely on a few large clients, do not necessitate broad public-facing quality assurance. Our system's performance is, therefore, a direct reflection of the digital visibility of the entities within the supply chain.

Furthermore, the fact that our method achieved over 20 percentage points higher accuracy compared to the non-fine-tuned LLM underscores the critical importance of embedding domain knowledge into the model. This finding suggests that for effective supply chain visibility, general linguistic capabilities are insufficient. Instead, inference must be grounded in specialized knowledge of component characteristics and industrial structures, which can only be achieved through domain-specific fine-tuning.

5.2 Practical Implication and Value

The comparison with the baseline procedures highlights the practical value of our approach. Manual supplier inquiries are still effective for Tier 1 suppliers; however, their reliance on direct communication constrains scalability and incurs substantial time costs. In contrast, our method attains comparable accuracy in a fraction of the time and, crucially, supports inference for Tier 2 suppliers and beyond—an area that traditional practices do not adequately cover. This capability positions the system as a key enabler of contemporary supply-chain-resilience initiatives.

We estimate that the proposed workflow can reduce the manual effort required for supply-chain mapping and investigation by as much as 80%. For example, verifying the provenance of 100 components typically obliges a procurement specialist to examine every

item individually. With our system, roughly 80 of those components can be resolved automatically with high confidence, allowing human expertise to be focused on the remaining 20 complex cases (e.g., single-source or safety-critical parts).

Interviews with POs indicate that an accuracy of approximately 60% is already considered operationally sufficient. POs recognize from experience that even exhaustive manual investigations rarely yield complete information: suppliers frequently relocate production for reasons such as cost optimization or business-continuity planning, rendering perfect traceability unattainable. Moreover, manual investigations are inherently slow. Semiconductor lifecycles, for instance, are short, and component configurations may change in less than two years; spending an entire month to locate the factory for a single product is no longer feasible. POs responsible for 40 products would need more than three years to cycle through all items at that pace, by which time many configurations would have changed again.

Consequently, POs accept a trade-off between accuracy and lead-time and regard 60% accuracy as an acceptable threshold, with the remaining 40% handled manually. Our method delivers at least 80% accuracy (Table 4), substantially exceeding the operational criterion established by POs and therefore meeting practical requirements.

5.3 Limitation and Future Work

Despite its promising results, this study has several limitations that present clear directions for future research. The primary limitation is the system's dependence on publicly available information. When public data is insufficient, our current process relies on external commercial company databases to supplement information. These databases are typically curated for financial risk assessment and often only contain registered head office locations, not specific factory sites. For SMEs, our system currently operates on the assumption that the headquarters and factory are co-located, which is a known limitation that can lead to inaccuracies.

To address these limitations, our future work will focus on two key areas: (a) Developing a Continuous Learning Feedback Loop: We plan to implement a human-in-the-loop system where procurement experts validate or correct the model's inferences. This labeled feedback will be systematically collected and used to continuously retrain and refine the model, improving its accuracy and domain-specific knowledge over time. (b) Enhancing Proactive Data Collection: To reduce reliance on the limitations of existing databases, we will explore methods for proactively collecting a wider array of public information. This includes developing sophisticated web-crawling techniques to gather data from a broader set of corporate websites, regional news reports, and industry-specific forums, thereby creating a richer dataset to overcome existing information gaps.

Future validation will focus on expanding the system's applicability to other industrial sectors. For example, plans are in place to apply the methodology to eight additional products in the electric industry and a few products in the food industry.

5.4 Ethical Consideration

The development of a technology capable of mapping deep-tier supply chains necessitates careful consideration of its ethical implications. A primary concern is the potential for misuse, such as improperly scouting a competitor's supply chain. To mitigate this risk, deployment of this technology must be governed by robust technical and policy-based safeguards. These include strict terms of use, access control mechanisms to ensure data is used only for its intended purpose, and the potential for abstracting output information to prevent overly granular or sensitive disclosures.

6 Conclusion

This paper addressed the critical challenge of achieving visibility into the deep tiers of complex global supply chains. We introduced a novel Manufacturing Site Inference Technology that leverages a fine-tuned Large Language Model to automatically identify specific factory locations from diverse and unstructured data. By processing disparate sources such as supplier lists and ISO certifications, our proposed method offers a transformative solution to the limitations of traditional, manual investigation techniques.

The empirical evaluation demonstrated the effectiveness of our approach. Validated on real-world procurement data, the system achieved over 80.1% accuracy in identifying correct manufacturing sites, substantially reducing the manual effort required for such granular analysis. In doing so, this work successfully delivered on its primary contributions: we developed a novel inference pipeline capable of navigating complex data; we established its robustness against the conflicting and incomplete information found in a real-world setting; and we confirmed its practical viability through high-accuracy performance. This research provides a tangible advancement in applying AI to enhance supply chain resilience through data-driven visibility.

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